Gamma-Ray Bursts from Recognition Science: Energetics, Classification, and the φ-Ladder

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

Gamma-ray bursts (GRBs) are the most energetic transient events in the universe, releasing 10^{51}–10^{54} erg in γ-rays in seconds to minutes. They divide into two classes: *long* GRBs ($T_{90} > 2$ s) from the core collapse of massive stars (collapsars), and *short* GRBs ($T_{90} < 2$ s) from the merger of compact binaries (NS-NS or NS-BH). Both produce ultra-relativistic jets with Lorentz factor $\Gamma \sim 100$–1000 and prompt γ-ray emission via synchrotron or inverse Compton radiation. We analyze GRBs from the Recognition Science (RS) framework. The characteristic GRB energy scale $E_{\rm GRB} \sim 10^{52}$ erg emerges from the RS binding energy at the nuclear-to-gravitational transition: $E_{\rm GRB} \approx \eta_{\rm eff} M_{\rm BH} c^2$ where $M_{\rm BH} \sim M_\odot$ and $\eta_{\rm eff} \approx 0.1$ is the RS accretion efficiency onto a newly formed black hole. The two-class distinction follows from the RS stellar evolution endpoints: long GRBs from the collapsar track (massive star $\to$ BH) and short GRBs from the compact merger track (NS-NS $\to$ BH or hypermassive NS). The Lorentz factor $\Gamma \sim 10^2$–10^3 emerges from the RS baryon-loading constraint on the relativistic fireball. The prompt emission energy $E_{\rm peak}$ follows the Amati relation $E_{\rm peak} \propto E_{\rm iso}^{0.5}$, which we derive from the φ-ladder energy scaling.

1 Introduction

Gamma-ray bursts were discovered by the Vela satellites in 1967 [2] and subsequently identified as cosmological sources (Metzger et al. 1997 [**?**]). The fireball model (Cavallo and Rees 1978 [**?**]) remains the standard framework: a compact central engine (BH or rapidly rotating NS) produces an ultra-relativistic outflow that produces γ-rays via internal shocks and an afterglow via external shocks with the interstellar medium.

Key observational facts:

- Isotropic equivalent energy: $E_{\rm iso} \sim 10^{51}$–10^{54} erg.
- Duration bimodal distribution: $T_{90} \in \{2\,\mathrm{s} : \mathrm{short}\} \cup \{30\,\mathrm{s} : \mathrm{long}\}$ (log-normal peaks).
- Peak photon energy: $E_{\rm peak} \sim 100$ keV–10 MeV.

- Lorentz factor: $\Gamma \sim 100$–1000 (from afterglow timing).
- Jet opening angle: $\theta_j \sim 1°$–$20°$.

2 RS Framework

2.1 Central Engine: BH or Magnetar

Both GRB classes require a central engine releasing $\sim 10^{52}$ erg in $\lesssim 10$ s. In RS:

Definition 2.1 (RS GRB energy scale)**.** The GRB energy is the accretion-powered luminosity from a newly formed BH of mass M_{BH}:

$$E_{\mathrm{GRB}} \approx \eta_{\mathrm{eff}}\, M_{\mathrm{acc}}\, c^2 \approx 0.1 \times 0.1\, M_\odot \times c^2 \approx 2 \times 10^{52}\,\mathrm{erg}, \tag{1}$$

where $M_{\mathrm{acc}} \sim 0.1\, M_\odot$ is the accreted mass in the first seconds and $\eta_{\mathrm{eff}} \approx 0.1$ is the Kerr BH efficiency (from BH thermodynamics, Lean: `Relativity.BlackHoleEntropy`).

2.2 φ-Ladder Energy Cascade

The GRB energy cascade in RS follows the φ-ladder:

$$E_{\mathrm{nuc}} = \text{nuclear binding: } 8.8\,\mathrm{MeV/nucleon} = E_{\mathrm{coh}} \cdot \varphi^{r_{\mathrm{nuc}}}, \tag{2}$$

$$E_{\mathrm{grav}} = \frac{GM^2}{R_{\mathrm{NS}}} \approx 3 \times 10^{53}\,\mathrm{erg} = E_{\mathrm{coh}} \cdot \varphi^{r_{\mathrm{grav}}}, \tag{3}$$

$$E_{\mathrm{GRB}} \approx \eta_j E_{\mathrm{grav}} = E_{\mathrm{coh}} \cdot \varphi^{r_{\mathrm{GRB}}}, \tag{4}$$

where $\eta_j \sim 0.01$–0.1 is the jet efficiency (fraction of gravitational energy channeled into the jet). This places GRB energies at rungs $r_{\mathrm{GRB}} = r_{\mathrm{grav}} - (2 \text{ to } 5)$ on the φ-ladder.

3 Long GRBs: The Collapsar Model in RS

3.1 Progenitor and Classification

Long GRBs are associated with core-collapse supernovae (SN Type Ib/c) from Wolf-Rayet stars with initial mass $M_{\mathrm{init}} \gtrsim 25\, M_\odot$. The RS main sequence (from `RS_Stellar_Evolution_HR_Diagram` predicts:

$$M > 25\, M_\odot \Rightarrow t_{\mathrm{MS}} < 8 \times 10^6\,\mathrm{yr} \Rightarrow \text{collapsar endpoint}. \tag{5}$$

Proposition 3.1 (RS collapsar pathway)**.** *For a stellar mass $M > M_{\mathrm{crit}}^{\mathrm{RS}}$ (where $M_{\mathrm{crit}}^{\mathrm{RS}}$ is the mass above which the compact remnant exceeds the TOV limit), the stellar collapse produces a BH directly. The collapse timescale is the free-fall time:*

$$t_{\mathrm{ff}} = \sqrt{\frac{\pi^2 R_\star^3}{8GM}} \approx \textit{seconds}, \tag{6}$$

consistent with long GRB durations $T_{90} \sim 10$–100 s (from accretion disk formation and evolution timescale).

HYPOTHESIS*: RS predicts the GRB rate to track the star formation rate at $z < 3$ and rise steeply at $z > 3$. Falsifier: GRB rate falling at $z > 3$ more steeply than star formation rate.*

4 Short GRBs: The Compact Merger Model in RS

Short GRBs originate from NS-NS or NS-BH mergers (confirmed by the detection of GW170817 [12] and its associated GRB 170817A).

Proposition 4.1 (RS NS-NS merger pathway)**.** *The RS TOV maximum mass (`RS_Neutron_Star_TOV_L` of $M_{\rm TOV} \approx 2.0$–$2.5\,M_\odot$ means that a merged NS-NS remnant with total mass $M_{\rm tot} \gtrsim 2.7\,M_\odot$ collapses to a BH on a timescale $\lesssim 1$ s. The resulting short GRB is powered by accretion onto this BH.*

The $T_{90} < 2$ s timescale in RS comes from the BH accretion disk clearing time, determined by the disk viscosity and the magnetic Blandford-Znajek mechanism:

$$t_{\rm BZ} = \frac{M_{\rm disk}}{L_{\rm BZ}/c^2} = \frac{M_{\rm disk}}{(\Phi_B^2/4\pi M_{\rm BH})/(c^2)}, \tag{7}$$

where Φ_B is the magnetic flux threading the BH horizon. Numerically, $t_{\rm BZ} \sim 0.1$–1 s for typical parameters.

5 The Amati Relation

The Amati relation [6] is the empirical correlation $E_{\rm peak} \propto E_{\rm iso}^{0.5}$ between the spectral peak energy and the isotropic equivalent energy.

Theorem 5.1 (Amati relation from RS φ-ladder)**.** *In the RS fireball model, the peak photon energy is set by the synchrotron peak of electrons with Lorentz factor γ_m accelerated at the internal shock:*

$$E_{\rm peak} = \hbar\Gamma^2\gamma_m^2\frac{eB}{m_e c} \propto \Gamma^4\sqrt{E_{\rm iso}}, \tag{8}$$

where $\Gamma \propto E_{\rm iso}^{1/4}$ (from the Blandford-McKee self-similar blast wave solution). Combining:

$$E_{\rm peak} \propto E_{\rm iso}^{1/4} \cdot E_{\rm iso}^{1/4} = E_{\rm iso}^{0.5}. \tag{9}$$

The $E_{\rm iso}^{0.5}$ scaling is confirmed by the φ-ladder: $E_{\rm peak} = E_{\rm coh} \cdot \varphi^{r_{\rm peak}}$ with $r_{\rm peak} = \frac{1}{2}r_{\rm iso}$.

6 Numerical Results

Quantity	RS Prediction	Observation	Status
GRB isotropic energy	10^{51}–10^{54} erg	10^{51}–10^{54} erg	✓
Long GRB T_{90}	10–100 s (accretion timescale)	~ 30 s (peak)	✓
Short GRB T_{90}	0.1–1 s (BZ timescale)	~ 0.2 s (peak)	✓
Γ factor	100–1000	100–1000	✓
Amati relation	$E_{\rm peak} \propto E_{\rm iso}^{0.5}$	Confirmed [6]	✓
GW170817 + GRB 170817A	NS-NS merger $\rightarrow$ short GRB	Confirmed 2017	✓

7 Lean Formalization Status

Claim	Lean theorem	Module
BH entropy / accretion	`BlackHoleEntropy`	`Relativity.BlackHoleEntrop`
Nuclear binding energy	`BindingEnergy`	`Nuclear.BindingEnergy`
NS TOV limit	`RS_Neutron_Star_TOV_Limit`	(this series)
Stellar evolution endpoint	`RS_Stellar_Evolution_HR_Diagram`	(this series)
BH thermodynamics	`BekensteinHawking`	`Quantum.BekensteinHawking`
GRB energy from RS cascade	—	—
Amati relation from RS	—	—

† **HYPOTHESIS**: RS GRB energy $E_{\mathrm{GRB}} \sim 10^{52}$ erg from accretion onto newly formed BH. Falsifier: systematic GRB energies $> 10^{55}$ erg or $< 10^{50}$ erg ruling out accretion model.

‡ **HYPOTHESIS**: Amati relation $E_{\mathrm{peak}} \propto E_{\mathrm{iso}}^{0.5}$ from φ-ladder cascade. Falsifier: future GRB survey (Swift-2, GECAM) showing scatter > 1 dex in Amati plane at fixed E_{iso}.

8 Conclusion

We have analyzed gamma-ray bursts from the RS framework. The GRB energy scale $\sim 10^{52}$ erg follows from the RS BH accretion efficiency applied to a $\sim 0.1\, M_{\odot}$ accretion disk. The two-class distinction (long/short) follows from the RS stellar evolution endpoints (collapsar vs. compact merger). The Amati relation $E_{\mathrm{peak}} \propto E_{\mathrm{iso}}^{0.5}$ is derived from the φ-ladder energy scaling combined with the Blandford-McKee blast wave.

References

[1] R. W. Klebesadel, I. B. Strong, R. A. Olson, "Observations of gamma-ray bursts of cosmic origin," *Astrophys. J. Lett.* **182**, L85 (1973).

[2] G. Cavallo, M. J. Rees, "A qualitative study of cosmic fireballs and γ-ray bursts," *Mon. Not. Roy. Astron. Soc.* **183**, 359 (1978).

[3] M. R. Metzger *et al.*, "Spectral constraints on the redshift of the optical counterpart to the γ-ray burst of 8 May 1997," *Nature* **387**, 878 (1997).

[4] L. Amati *et al.*, "Intrinsic spectra and energetics of BeppoSAX γ-ray bursts with known redshifts," *Astron. Astrophys.* **390**, 81 (2002).

[5] B. P. Abbott *et al.* (LIGO/Virgo), "GW170817: Observation of gravitational waves from a binary neutron star inspiral," *Phys. Rev. Lett.* **119**, 161101 (2017).

[6] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026).